ARNAV SHARMA

Computer Science and Engineering

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Computer Science Engineer with experience in **distributed systems, edge, AI, & IoT**. Proficient in **Python**, with course work in **C, Go, Grafana, cloud, and CI/CD**. Strong problem-solving background with notable competitive programming achievements. Beyond tech, a **FIDE chess** player and **table tennis** hobbyist. Skilled in Agile teamwork and management.

Experience

Akamai Technologies – Intern (Bengaluru, Karnataka) **Jun 2026 – Aug 2026**

- Worked with Edge Servers Team, a core component of the CDN infrastructure at the **world's largest CDN** provider, contributing to projects including the **GHost Cache System Repository** serving Akamai's edge network, responsible for 35% of total global web traffic. Built an Edge Log Monitoring System using Grafana Loki, LogQL and Alloy.
- Engineered **low-level production replay systems** in Python, JavaScript & Go to analyse cache hits and misses from production logs at Edge Servers and replay traffic to detect bugs at edge, improve performance and reduce latency.
- Collaborated with a globally distributed agile team using tools like Jira, bitbucket etc. Authored 3 Confluence documents.

iMerit Technology Services Pvt. Ltd. – Technical Consultant (Remote | IN) **Nov 2025 – Jan 2026**

- Freelance consultant collaborating within the global iMerit Scholars network on AI evaluation projects.
- Worked on the Orion Project, evaluating multimodal AI image outputs for prompt alignment and logical consistency.

Projects

Production AccessLog Replay Orchestration Tool for CDN *(Python, AWS, k6, JavaScript)* **Aug 2024 - Present**

- **Modular Replay Pipeline:** Architected a 4-component Python pipeline to parse and normalize gigabyte-scale production logs (r-lines & f-lines) while preserving exact temporal ordering and edge-routing behaviors.
- **High-Fidelity Replay Driver:** Programmed a k6 load-testing driver executing concurrent Virtual Users to validate HTTP status codes, cache hit/miss paths, and origin turnaround times.
- **Custom Edge Infrastructure:** Replayed traffic against a custom-built CDN hosted on AWS EC2 instances configured as edge servers to benchmark real-world edge routing and caching performance.

https://github.com/ArnavSharma2908/replay-engine-edge-servers

Mystery Animal Voice Game *(ESP32, Router, INMP441 I2S Microphone, MicroPython)* **Sep 2024 - Oct 2024**

- Created an ESP32-powered Mystery Animal guessing game for Robotics Club@NIE, where users guess the simulated animal by asking a maximum of 20 yes/no questions using an INMP441 I2S microphone and Python backend.
- Designed an end-to-end pipeline where user questions are processed on a Python server, evaluated against mystery-animal logic, and a response is returned to drive red/green LED indicators and buzzer-based audio feedback.
- Core game logic and interaction flow were adapted and derived from Google's Mystery Animal experiment (2017).

https://github.com/googlecreativelab/mystery-animal

Delhi Metro Transit Corporation and ChatBot *(Python, matplotlib, pyttsx, WeatherAPI, SQL)* **Oct 2022 - Jan 2023**

- Developed a Python-based metro guide providing route details, fare estimates, and travel times across about 230 stations using Matplotlib for mapping and MySQL for data storage.
- Incorporated METROBOT, an AI-scripted chatbot, for user queries, FAQs, and assistance.
- Added real-time news, weather information along with in-chat Wikipedia access in Metrobot.

https://github.com/ArnavSharma2908/Delhi-Metro-Transit-Corporation-and-Chatbot

Education

The National Institute of Engineering, Mysuru **Sep 2023 - Sep 2027**
Bachelor of Engineering in Computer Science and Engineering

Salwan Public School, Mayur Vihar, Delhi **Apr 2021 - Mar 2023**
High School Certificate in Science with CS

Achievements

ICPC Regionalist – 2025 Asia West Amritapuri Multisite Regional Contest **Nov 2025**
1st Position at Brainiacs Track, Codeft 5.0 Hackathon, Army Institute of Technology, Pune **Apr 2025**
2nd Position at National Competitive Programming Potpourri, Shaastra TechFest at IIT, Madras **Jan 2025**
3rd Position at Inter-Salwan Computer Science Investigatory Project Expo, SPSMV **Nov 2022**

Core Skills

Languages and Development Tools: Python, Java, GoLang, C, HTML, CSS, JavaScript, LogQL
Tools: Unix, Version Control System (Git), REST APIs, Jira, Grafana Loki, k6,
Databases: SQL/NoSQL, MySQL, PostgreSQL, MongoDB
Libraries: Matplotlib, NumPy, LangChain, BeautifulSoup, Pyttsx3, FastAPI, Node.js, Express.js
DevOps: CI/CD, GitHub Actions, Docker, AWS (EC2, S3, EBS, Amplify)

Interests & Activities

Chess: FIDE rating – 1411; Delhi Zonal Chess Runner-up, DOE; VTU Intercollegiate Chess Winning Team (ROB Division)
Contributions: Open-source contributor at GSSOC, spyder, Hacktober; Tech Lead at IEEE Computer Society; Robotics Club; Event Volunteer, Smart India Hackathon – Hardware Nodal Centre; Co-Lead, Kneights Templar chess community